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Social motivations of live-streaming viewer engagement on Twitch

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ABSTRACT

Little is known about the motivations underlying viewer engagement in the rapidly growing live-streaming multimedia phenomenon. This study trialled an eight-factor socio-motivational model, based on Uses and Gratifications Theory, to explain four aspects of live-stream viewer engagement. Cross-sectional data was collected through an international, online self-report survey of Twitch users ($N = 2227$). Multiple and ordinal linear regression analyses identified six motivations which helped to explain live-stream engagement: social interaction, sense of community, meeting new people, entertainment, information seeking, and a lack of external support in real life. Compared to mass media, viewer motivations to engage in live-stream entertainment appear to have a stronger social and community basis. Furthermore, live-stream viewers who preferred smaller channels (<500 viewers) were more motivated by social engagement than viewers who preferred larger channels. These findings offer insight into the motivations for live-stream engagement, and help to lay a foundation for further research.

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1. Introduction

Live-streaming is an interactive form of internet-based multimedia entertainment that has grown rapidly in popularity worldwide since 2011 (Needleman, 2015; Twitch, 2017). Live-streaming has become so popular that, in some cases, there are more people watching others do activities, such as play computer games, than doing the activity themselves (Kaytoue, Silva, Cerf, Meira Jr, & Raïssi, 2012). Unlike previous streaming services, such as television and YouTube, live-streaming offers real-time human interaction between the streamer and viewers, facilitating their ability to interact with each other. While online technology has made this development possible, what is driving the rapid growth in live-streams, is not well understood from a psychological perspective. Despite the popularity of live-streaming consumption, little psychological research has been conducted. Some past research has explored live-streaming consumption motivations including tension release, escapism, and acquiring knowledge (e.g., Sjöblom & Hamari, 2017; Sjöblom, Törhönen, Hamari, & Macey, 2017), however social-based motivations such as needs for interaction, connection, and community, have not been studied within the context of live-streaming. This study seeks to address this gap by exploring the relationship between live-stream viewer socio-motivations and their psychological and behavioural live-stream engagement.

1.1. Live-streaming

Live-streaming is a rapidly growing internet-age phenomenon. Nearly 10 million people tuned in to Twitch per day in 2016 to stream, watch, and interact with others (Twitch, 2016) Twitch is currently the largest live-streaming platform (Twitch, 2016). Live-streamers broadcast themselves playing video games, eating food, painting, dancing, and so on, in real time, to a public audience. Viewers watch and listen to the streamer, but are also able to directly interact with, and responding to, what the streamer says and does. This facilitates a two-way connection where streamers

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can directly acknowledge and respond to viewers, and viewers can actively participate in, and influence, live-stream broadcasts (Sjöblom & Hamari, 2017).

Twitch has grown rapidly in popularity. For example, Twitch exceeded Facebook in peak internet traffic in 2014 (Fitzgerald & Wakabayashi, 2014; Gamezone, 2014), with monthly viewers doubling from 45 million in December 2013 to 100 million in December 2014 (Leslie, 2015). Other sites such as YouTube, Dailymotion, and Facebook have introduced live-streaming services with similar features, however, Twitch is currently the most popular (D'onfro, 2015; O'Neill, 2014; Wilhelm, 2013). Live-streaming began as a niche, gaming-oriented domain, but is diversifying and growing into a broader social media trend. Despite its rapidly growing trajectory, there is limited research about streamer and viewer motivations.

1.2. Uses and Gratification Theory

Uses and Gratification Theory (UGT) evolved from theories about why people choose different types of media consumption (Rubin, 2009), and has been used to analyse consumer engagement in online social networking sites (Ku, Chen, & Zhang, 2013; West & Turner, 2010; Whiting & Williams, 2013), video sharing (Chiang & Hsiao, 2015), live-streaming (Sjöblom & Hamari, 2017), and eSports (Hamari & Sjöblom, 2017). According to UGT, people actively seek out, consume, and participate in media that fulfil their individual needs and match their preferences better than other media options (Ruggiero, 2000; Shao, 2009). UGT offers a functional approach which can help to understand how different forms of media engagement relate to the fulfilment of various psychological needs. Thus, the UGT provides a framework for understanding consumer choices and roles in media engagement as behaviours which are aimed at the fulfilment of individual psychological needs.

1.3. A socio-motivational model of live-stream participation

Building on UGT, theory and research about social media consumption suggests at least eight socio-motivations for live-stream engagement: entertainment, information seeking, meeting new people, social interactions, social support, sense of community, social anxiety, and external support. These motivations are explained in the following sections.

To capture live-stream engagement across various domains, four indicators of live-stream engagement were conceptualised. Emotional connectedness to the live-stream platform as an indicator of psychological engagement, time spent (per week) on the live-stream platform as a behavioural indicator of engagement. Time subscribed (in months) to channels on the live-stream platform and amount donated (in USD) to channels on the live-stream platform as financial indicators of engagement. Fig. 1 provides a visual representation of the proposed motivators of live-stream engagement and indicators of engagement.

1.3.1. Entertainment and information seeking

Entertainment and information seeking have been previously identified as important motivators for consuming streaming services (Cheung & Huang, 2011; Hamilton, Garretson, & Kerne, 2014; Sjöblom & Hamari, 2017). An important aspect of streaming is to provide an entertaining media experience. The most popular Twitch live-streams host eSport tournaments and professional gamer streams which allow viewers to watch and learn from the best players in the world (Cheung & Huang, 2011). Live-streams also commonly conduct product reviews either intentionally or by showing games being played. Therefore, streams are also used to provide previews of gameplay, allowing potential buyers to ask questions or request opinions of the game. Thus, entertainment and information seeking are proposed as two key reasons for live-stream engagement.

1.3.2. Meeting new people, social interactions, and sense of community

Social motivations, such as meeting new people, social interaction, and sense of community are important to live-stream viewers. Hamilton et al. (2014) found that live-streams serve as virtual “third places” where communities form and grow, with viewers using chat rooms to converse, laugh, and joke with each other about content they are watching. Chat rooms foster sociability, with regular viewers and moderators playing specific roles in welcoming and engaging newcomers (Hamilton et al., 2014). Thus, live-stream environments facilitate meeting new people, a common motivator for participation in online communities (Brandtzæg & Heim, 2009; Ridings & Gefen, 2004; Zhu & Chang, 2014).

A sense of community is also important in live-streaming (Hamilton et al., 2014). Sense of community includes experiences of membership, influence, fulfilment of needs, and belonging to the channel, streamer, and other participants (McMillan & Chavis, 1986). This suggests that live-stream viewers are attracted to channels where they feel noticed, important, and influential. Live-stream viewers can have their membership needs fulfilled by participating in community success and sociability. Shared experiences and continued participation in live-streams helps to develop connectedness with other stream participants and fosters a sense of community in the channel.

1.3.3. External support, social support, and social anxiety

People may participate in online communities to compensate for a lack of community in real life (Miller, 2011). Online communities can be particularly beneficial for the psychological well-being of participants who lack external support from their family, friends, and local community (Bargh & McKenna, 2004). Online social interactions have been reported to reduce loneliness by presenting low-risk opportunities for self-disclosure, online social support, and involvement (Valkenburg & Peter, 2009).

Online interaction can also help individuals with social anxiety, who find it hard to socially engage with others in real life (Baumeister & Leary, 1995; Desjarlais & Willoughby, 2010; Mazalin & Klein, 2008). Live-stream environments can provide low-threat alternatives to real life socialising, removing barriers socially anxious individuals may experience in real life.

1.4. Indicators of live-stream engagement

This research seeks to capture live-stream engagement on four factors: emotional connectedness to the service (psychological attachment), amount of time watching live-streams, number of channel subscriptions, and amount of money donated (behavioural attachments).

Emotional Connectedness. Emotional connectedness refers to how psychologically attached the viewer is to the live-stream platform and specific live-stream communities. Emotional connectedness to social media platforms, such as Facebook, has been explored (Ellison, Steinfield, & Lampe, 2007), however emotional connectedness to live-stream platforms has not been previously researched.

Time Spent. A second indicator of live-stream engagement is time spent watching live-streams. Twitch viewers reported

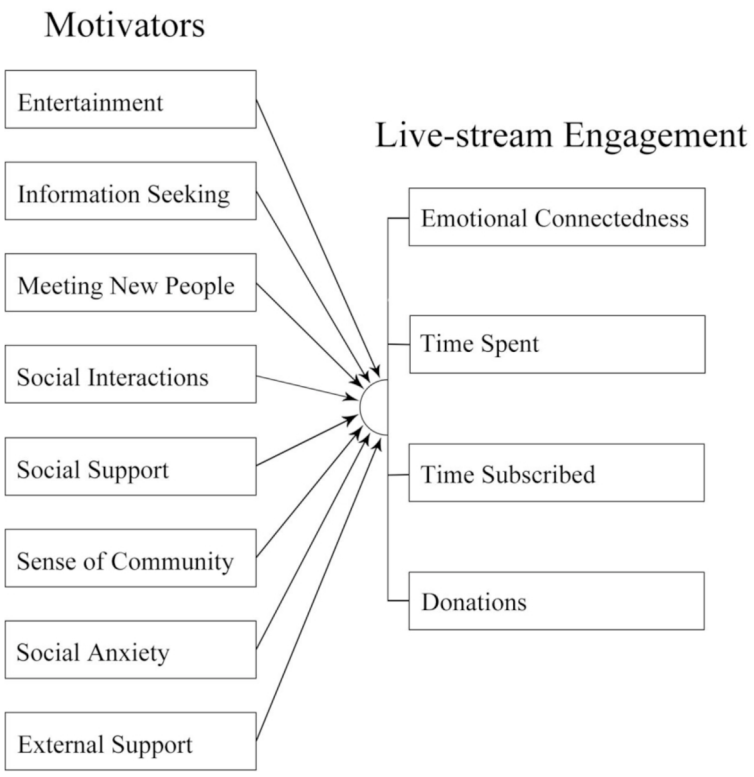


Fig. 1. Proposed motivators of psychological, behavioural, and financial live-stream engagement.

spending an average 106 min every day watching live-streams and nearly half of the surveyed users reported spending over 20 h per week on Twitch, equivalent to a half-time job (Twitch, 2016).

Time Subscribed. A third indicator of live-stream engagement is subscription. Subscribing involves direct financial investment to support a streamer's channel. Subscribing to a Twitch channel costs the viewer USD 4.99 per month. In return, subscribers receive a badge next to their name, and use of customised emoticons to express feelings or intended tone. Subscribing often provides other social-oriented benefits, such as access to a live-stream voice server to engage verbally with other subscribers and invitations to participate in games or activities being broadcasted by the streamer. Viewers tend to subscribe for social integration reasons, essentially to enhance connections with the channel community (Sjöblom & Hamari, 2017).

Donations. A fourth indicator of live-stream engagement is financial donation to the streamer (via PayPal) or charities (Twitch, 2017). Unlike subscriptions, donations do not provide access to additional content. Instead, the donator is usually thanked personally by the streamer, celebrated in the chat room, and may have their name displayed on the stream as the “Top Donator” or a “Recent Donator”.

1.5. Motivations across different sizes of live-stream environment

Some viewers prefer small streams, as they facilitate meaningful social interactions that are harder to achieve in large, fast moving chat rooms (Hamilton et al., 2014). Streamers have also reported that they can interact effectively with only 100 to 150 participants and that with more people the personal interaction between the streamer and the viewer breaks down (Hamilton et al., 2014). This is consistent with the cognitive limit on the number of meaningful social relationships that can be sustained, as suggested by Dunbar (1992). Smaller channels may therefore provide viewers with greater individual attention, participation, and influence in content; factors that increase overall belonging and community (Hamilton et al., 2014; McMillan & Chavis, 1986). UGT suggests that viewers will actively seek and choose media environments that fulfil their individual psychological needs (Ruggiero, 2000; Shao, 2009). From this perspective, viewers who prefer smaller channels are likely to be more driven by social needs than viewers who prefer larger channels. Viewers who prefer larger channels are more likely to be motivated by less social needs, such as entertainment and information seeking. However, this has not been empirically tested.

1.6. The present study

This study investigates what motivates people to engage in live-streams on the internet, based on a UGT framework. More specifically, the current study investigates the extent to which eight socio-motivators (entertainment, information seeking, meeting new people, social interaction, social support, sense of community, social anxiety, and external support) explain four types of Twitch live-stream engagement (emotional connectedness, time, subscriptions, and donations). The study also aims to investigate whether relationship between live-stream viewer motivations and indicators of engagement vary by channel sizes, as has been suggested by past research applying a UGT framework (Hamilton et al., 2014). Accordingly, two hypotheses are proposed.

Firstly, it is hypothesised that entertainment, information seeking, meeting new people, social interaction, sense of community, social support, social anxiety, and a lack of external support, will explain live-stream engagement. Secondly, it is hypothesised that social motivators (social interaction, sense of community, social support, meeting new people, social anxiety and external support) will better explain live-stream engagement behaviours for viewers who prefer small channels than for viewers who prefer larger channels.

2. Method

2.1. Participants

There were 3611 responses to an online survey, however 38.3% of these responses were removed because they reported typically watching Twitch for 0 h per week, missed 50% or more of the survey questions, provided unrealistic answers (e.g., watching for more hours than possible in a week), or finished the survey in under 5 min, leading to a final sample of 2227 participants. The main referral sources were Reddit ($n = 1949$; 87.5%), direct referrals ($n = 91$; 4.1%), Twitch ($n = 77$; 3.5%), Twitter ($n = 70$; 3.1%), and Facebook and other sources ($n = 40$; 1.8%). Participant ages ranged between 16 and 49 years ($Mdn = 21$, $SD = 4.62$). There were 2129 males (95.6%), 86 females (3.9%), 11 who reported their gender as other (0.5%), and one participant who did not report their gender (0.0%).

2.2. Measures

This study used an online, self-report survey which consisted of 79 items (70 closed-ended items and 9 open-ended items). Each section of the survey is described in the following sections.

Emotional Connectedness. A five-item Emotional Connectedness scale was adapted from the Facebook Intensity Scale by replacing "Facebook" with "Twitch". The Facebook Intensity Scale was designed to measure emotional connectedness to Facebook and daily integration of Facebook use (Ellison et al., 2007) and was scored on a five-point Likert scale (ranging from 1 = strongly disagree to 5 = strongly agree).

Time Spent, Time Subscribed, and Donations. The extent of live-stream behavioural engagement was assessed using three single-item open-ended questions: "On average, how many hours do you spend watching Twitch per week?", "How many months in total have you been subscribed to a Channel/Stream on Twitch?", and "Estimate how much money (USD) you have donated in total to broadcasters on Twitch?".

Channel Size Preference. Participants were asked to estimate the number of viewers who are typically in the Twitch channel/s which they spend the most time viewing. This variable was recoded into three ordinal categories: Small (<500 viewers), Large (500–10,000 viewers), and Very Large (>10,000 viewers).

Entertainment. Entertainment motivation for participating in live-streaming was measured by adapting three items from Chang and Zhu (2011). The items were all positively worded and scored on a five-point Likert scale (ranging from 1 = strongly disagree to 5 = strongly agree), as were all the other items adapted from Chang and Zhu (2011) for the current study which are described in the following sections. To adapt each of these Chang and Zhu (2011) items for the current study, "social networking sites" was replaced with "Twitch". Cronbach's alpha for Entertainment in the current study was low ($\alpha = .55$) compared to past research ($\alpha = .76$; Chang & Zhu, 2011). Consequently, the item "I use Twitch to pass the time" was removed, raising the Cronbach alpha to .61.

Information Seeking. Information seeking and learning motivation for participating in live-streaming was measured using four items adapted from Chang and Zhu (2011).

Meeting New People. Friend seeking motivation for participating in live-streams was measured using three items adapted from Chang and Zhu (2011).

Social Interactions. This measure was based on four items about social interaction motivations to participate in virtual communities (Chiu, Hsu, & Wang, 2006). The items were adapted by replacing "virtual community" with "Twitch". The scale was scored on a five-point Likert scale (ranging from 1 = strongly disagree to 5 = strongly agree).

Social Support. This measure was based on four items about seeking social support motivations to use social networking sites (Kim, Sohn, & Choi, 2011). The items were adapted by replacing "social networking sites" with "Twitch". Each item was scored on a five-point Likert scale (ranging from 1 = strongly disagree to 5 = strongly agree).

Sense of Community. This measure was based on eight items from the Brief Sense of Community Scale (McMillan & Chavis, 1986; Peterson, Speer, & McMillan, 2008). The items were adapted by substituting "community" with "stream" or "channel". Participants were asked to answer the items with their favourite stream in mind. The scale was scored on a five-point Likert scale (ranging from 1 = strongly disagree to 5 = strongly agree).

Social Anxiety. This measure consisted of 15 items about individual's distress when initiating and maintaining conversations in real life (Mattick & Clarke, 1998). While the original measure contained 20 items, five items were excluded due to poor loading with other items, not uncommonly observed in other research (Rodebaugh, Woods, & Heimberg, 2007). Participants rate each item on a five-point scale (ranging from 0 = not at all characteristic or true of me to 4 = extremely characteristic or true of me). Responses were summed, and ranged from 0 to 60, with higher scores reflecting higher levels of social anxiety.

External Support. This measure consisted of 11 items from the Belonging subscale of the Interpersonal Support Evaluation List (Cohen & Hoberman, 1983). This subscale measures respondents' perceived availability of friends or social networks that they can spend time with in real life. Respondents are asked to rate their agreement with each item (0 = Mostly false or 1 = Mostly true). After reverse-scoring five negatively worded items, responses were summed, with higher scores reflecting higher perceived belongingness to a social network in real life.

2.3. Procedure

A cross-sectional, non-experimental design was used. Participants aged 16 years and over were recruited worldwide by advertising online on Reddit, Twitter, Facebook, Twitch, and direct online contact. Participation was voluntary. Participants were offered the incentive of taking part in a draw of two USD 50 gift cards to the online game store Steam. Data was gathered using an anonymous online survey over a two-week period during May, 2016. The survey took approximately 10–20 min to complete ($M = 19{:}35$, $Mdn = 9{:}34$). Item sets were randomized in order to mitigate order effects and reduce the potential for response sets. The study received ethics approval from the University of Canberra Human Research Ethics Committee (#16–84).

2.4. Validity and reliability

Internal consistency was assessed through the Cronbach's Alpha, CR and AVE measures. All indicators exceeded the recommended thresholds (Fornell & Larcker, 1981; Nunnally, 1967), and thus we can conclude that internal consistency for the model was met. Discriminant validity was also met, with the square root of the AVE for each of the constructs being higher than the correlation for any other construct (Chin, 1998; Fornell & Larcker, 1981; Jöreskog & Sörbom, 1996). The results of these validity calculations are displayed in Table 1.

3. Results

3.1. Descriptive statistics

There was less than 5% data for missing for each item. Missing

Table 1
Cronbach's alpha, composite reliability, average variance extracted and fornell-larcker criterion for the eight socio-motivators and four indicators of live-stream engagement.

Variable	Cronbach's alpha	CR	AVE	1	2	3	4	5	6	7	8	9	10	11	12
Live-stream Engagement															
1. Emotional Connectedness	0.781	0.848	0.529	0.727											
2. Time Spent	1.000	1.000	1.000	0.485	1.000										
3. Time Subscribed	1.000	1.000	1.000	0.240	0.187	1.000									
4. Donations	1.000	1.000	1.000	0.288	0.190	0.567	1.000								
Socio-motivators															
5. Entertainment	0.617	0.839	0.723	0.291	0.166	0.044	0.065	0.850							
6. Information Seeking	0.781	0.859	0.605	0.369	0.133	0.073	0.098	0.154	0.778						
7. Meeting New People	0.864	0.917	0.786	0.474	0.184	0.307	0.355	0.083	0.348	0.887					
8. Social Interactions	0.887	0.922	0.748	0.451	0.215	0.371	0.448	0.069	0.242	0.758	0.865				
9. Sense of Community	0.860	0.890	0.514	0.511	0.152	0.290	0.305	0.175	0.334	0.532	0.552	0.717			
10. Social Support	0.813	0.875	0.637	0.386	0.145	0.244	0.285	0.031	0.304	0.677	0.614	0.458	0.798		
11. External Support	1.000	1.000	1.000	−0.037	−0.137	−0.010	−0.007	−0.004	0.057	0.006	0.019	0.031	−0.041	1.000	
12. Social Anxiety	0.929	0.938	0.504	0.119	0.092	0.003	−0.021	0.035	0.091	0.078	0.036	0.090	0.127	−0.487	0.710

value analysis using Little MCAR's test was non-significant, suggesting that data was missing completely at random and thus listwise deletion was appropriate. An exploratory factorial analysis with one factor (no rotation) revealed that the variance explained by a single factor was less than 50% (32.67%), suggesting the data was free from common method bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Due to positive skewness of Time Spent, Time Subscribed, and Donations, these variables were recoded into ordinal data. Groupings of ordinal variables are presented in Table 2 and descriptive statistics for socio-motivators in Table 3.

3.2. Socio-motivational explanation of emotional connectedness to Twitch

To test how well the socio-motivational model explained Emotional Connectedness to Twitch, multiple linear regression (MLR) was used. Normal probability plot and scatterplot of the standardised residuals indicated that assumptions of normality, homoscedasticity, and linearity of residuals were met. Collinearity was acceptable (VIF < 3). There were 26 cases which exceeded the maximum Mahalanobis distance of 26.12 ($df = 8$, $\alpha < .001$). As the presence of these cases did not change the conclusions, they were retained for analysis.

The MLR accounted for 36.3% of variance in Emotional Connectedness to Twitch ($F(8, 2179) = 154.91$, $p < .001$). Regression coefficients and confidence intervals (CIs) are displayed in Table 4. In descending order, the most variance was positively explained by Sense of Community, Entertainment, Information Seeking, Social Interactions, and Meeting New People.

3.3. Socio-motivational explanation of time spent watching Twitch

To test how well the socio-motivational model explained Time watching Twitch, Ordinal Linear Regression (OLR) was used. The OLR provided a good fit to the data (Pearson Goodness-of-Fit test, $\chi^2 = 6548.71$, $df = 6553$, $p = .513$) and explained Time watched over and above the alternate intercept model ($\chi^2 = 216.09$, $df = 8$, $p < .001$), accounting for 9.4% (Cox and Snell Pseudo R^2) of the variance in Time watched. Regression coefficients and confidence intervals are shown in Table 5. In descending order, the most variance was positively explained by Entertainment, Social Interactions, and a lack of External Support.

Table 3
Descriptive statistics for socio-motivation variables for Twitch engagement.

Variable	*M*	*SD*	Skewness	Kurtosis
Entertainment	4.61	0.51	−1.35	1.99
Information Seeking	3.05	1.03	−0.28	−0.82
Meeting New People	1.74	1.00	1.34	0.89
Social Interactions	1.85	1.10	1.27	0.55
Sense of Community	3.08	0.90	−0.13	−0.38
Social Support	1.44	0.69	1.92	3.79
External Support	5.31	3.02	−0.01	−0.99
Social Anxiety	19.28	13.30	.65	−0.24

Table 4
Socio-motivational MLR coefficients for explaining emotional connectedness to Twitch.

Variable	*B* (99.9% CI)	β	sr^2
Sense of Community	.24 [.17, .30]	.23	.04*
Entertainment	.38 [.28, .48]	.20	.05*
Information Seeking	.14 [.09, .19]	.15	.02*
Social Interactions	.10 [.03, .17]	.15	.01*
Meeting New People	.11 [.02, .19]	.14	.01*
External Support	-.01 [-.03, .01]	-.05	.00
Social Anxiety	.00 [-.00, .01]	.04	.00
Social Support	.03 [-.07, .13]	.02	.00

Note. *$p < .001$.

3.4. Socio-motivational explanation of subscription to Twitch

To test how well the socio-motivational model explained Time subscribed to Twitch, Ordinal Linear Regression (OLR) was used. An OLR model provided a good fit to the data (Pearson Goodness-of-Fit

Table 2
Descriptive statistics for Twitch engagement and channel size preference.

Channel Size (viewer count)			Time Subscribed (months)			Time Spent (hours per week)			Donations (USD)		
Range	% of total	*n*	Range	% of total	*n*	Range	% of total	*n*	Range	% of total	*n*
Small (<500)	19.2	487	0	61.7	1375	1–5	25.4	565	0	71.3	1588
Large (500–10,000)	19.1	1029	1–7	19.1	425	6–10	27.2	607	1–10	9.6	214
Very large (>10,000)	61.8	711	7+	19.2	427	11–20	25.8	573	11–50	10.7	239
						>21	21.6	482	51–150	4.3	95
									>150	4.1	91

Table 5
Socio-motivational OLR coefficients for explaining time watching Twitch.

Variables	Estimate (99.9% CI)	*SE*
Entertainment	.52* [.26, .78]	.08
Social Interactions	.31* [.13, .51]	.06
External Support	-.09* [-.14, −.04]	.02
Information Seeking	.13 [-.01, .27]	.04
Social Support	-.04 [-.30, .21]	.08
Sense of Community	.03 [-.15, .20]	.05
Meeting New People	.03 [-.19, .25]	.07
Social Anxiety	.00 [-.01, .01]	.00

Note. *SE* = Standard Error; *$p < .001$.

Table 7
Socio-motivational OLR coefficients for explaining Twitch donations.

Variables	Estimate (99.9% CI)	*SE*
Social Interactions	.66* [.45, .87]	.06
Sense of Community	.29* [.06, .52]	.07
Entertainment	.23 [-.11, .57]	.10
Information Seeking	-.08 [-.26, .10]	.05
Meeting New People	.05 [-.20, .30]	.08
External Support	-.04 [-.10, .02]	.02
Social Support	.03 [-.26, .31]	.09
Social Anxiety	-.01 [-.02, .01]	.01

Note. *$p < .001$.

test, $\chi^2 = 4378.68$, $df = 4366$, $p = .443$) and explained Subscriptions over and above the alternate intercept model ($\chi^2 = 317.210$, $df = 8$, $p < .001$), for 13.5% of variance in Twitch Subscription behaviour. Ordinal regression coefficients and confidence intervals are displayed in Table 6. In descending order, the most variance was positively explained by Social Interactions and Sense of Community.

3.5. Socio-motivational explanation of donations to Twitch

To test how well the socio-motivational model explained Donations to live-streams, Ordinal Linear Regression (OLR) was used. The OLR model provided a good fit to the data (Pearson Goodness-of-Fit test, $\chi^2 = 8165.70$, $df = 8740$, $p = 1.000$), and explained Donations over and above the alternate intercept model ($\chi^2 = 399.91$, $df = 8$, $p < .001$), accounting for 16.7% of variance in Donations. Regression coefficients and confidence intervals are shown in Table 7. In descending order, the most variance was positively explained by Social Interactions and Sense of Community.

3.6. Socio-motivational explanation of Twitch engagement: summary

A summary of the MLR and OLR findings for relationships between the eight socio-motivators and four indicators of Twitch live-stream engagement is presented in Table 8. The socio-motivational model significantly explained live-stream engagement for each measures of engagement: Emotional Connectedness (37.1%), Time (9.4%), Subscriptions (13.9%), and Donations (16.8%). Motivations varied in significance and strength and across the type of live-stream engagement. Overall, the main motivators were Social Interactions and Sense of Community. Social Support and Social Anxiety were the only motivators that did not explain any type of engagement.

3.7. Socio-motivational explanation of live-stream engagement by channel size

Using a series of nine OLRs, the socio-motivational model was used to explain the three measures of Twitch across three preferred channel sizes (Small, Large, and Very Large; see Table 9). Emotional Connectedness was not included because attachment to the overall platform was not hypothesised to differ by channel size preference.

The utility of the socio-motivational model was greater with participants who had smaller channel size preferences, with nearly double the variance in engagement of Small channels explained by the socio-motivational predictors than for Large of Very Large streams.

Table 6
Socio-motivational OLR coefficients for explaining Twitch subscriptions.

Variables	Estimate (99.9% CI)	*SE*
Social Interactions	.49* [.29, .69]	.06
Sense of Community	.31* [.10, .52]	.06
Information Seeking	-.12 [-.28. .04]	.05
Meeting New People	.10 [-.14, .34]	.07
Entertainment	.08 [-.22, .39]	.09
External Support	-.03 [-.08, .03]	.02
Social Support	.02 [-.26, .30]	.09
Social Anxiety	-.01 [-.02, .01]	.00

Note. *SE* = Standard Error; *$p < .001$.

4. Discussion

This study examined viewer motivations for engaging in Twitch live-stream communities. It sought to identify factors that explain psychological and behavioural live-stream engagement and whether such motivations vary per channel size.

The utility of the proposed socio-motivational model was partially supported. Six of the eight motivators (social interaction, sense of community, meeting new people, entertainment, information seeking, and external support) significantly explained at least one indicator of live-stream engagement (emotional connectedness, watching, subscribing or donating). Social support and social anxiety were the only two motivators that were not associated with any type of live-stream engagement. This model explained twice as much of the engagement of participants who preferred smaller sized channels over larger channels.

The socio-motivational model most strongly explained emotional connectedness to Twitch, although this was likely partly due to emotional connectedness being the only parametric engagement measure which allowed for use of MLR. Emotional connectedness was best explained by the social dynamics of the live-stream environment (i.e., sense of community, social interaction, and meeting new people), as well as by a desire for entertainment and information seeking. While this research used emotional connectedness to measure psychological engagement, future research may consider utilising this measure as a predictor for behavioural live-stream participation. If an individual is emotionally attached to Twitch, he or she may be more motivated to watch, subscribe, and donate.

The amount of time viewers spent engaged in Twitch live-streams was significantly explained by entertainment, information seeking and social interaction, but also by a lack of external support. Entertainment and information seeking factors were unsurprising, as a desire of entertainment and learning would intuitively lead to increased time spent watching. These two factors have also been identified as motivators to watch live-streams in previous research (Chen & Lin, 2018; Hamilton et al., 2014; Sjöblom & Hamari, 2017; Sjöblom et al., 2017), as well as other types of online media related to this field, such as social media (Papacharissi & Mendelson, 2010), video services (Cha, 2014; Hanson & Haridakis, 2008) and esports (Cheung & Huang, 2011; Hamari & Sjöblom, 2017). The addition of social interactions is insightful, providing more evidence of social motivations to engage in live-

Table 8
Summary of significant motivators of each type of live-stream engagement.

	Emotional Connectedness	Time	Subscriptions	Donations
Entertainment	+*	+*		
Information Seeking	+*			
Meeting New People	+*			
Social Interactions	+*	+*	+*	+*
Sense of Community	+*		+*	+*
Social Support				
External Support		-*		
Social Anxiety				
Variance accounted for	36.3%	9.4%	13.5%	16.7%

Note. +/− indicates whether effect is positive or negative; *$p < .001$.

Table 9
Summary of significant motivations of each type of live-stream engagement of each channel size.

	Time Spent			Time Subscribed			Donations		
	Small	Large	Very large	Small	Large	Very large	Small	Large	Very large
Entertainment	+**		+**						
Information Seeking	+*			-*					
Meeting New People									
Social Interactions	+**	+**	+*	+*	+**	+**	+**	+**	+*
Sense of Community					+**				
Social Support									
External Support	-*	-**	-*						
Social Anxiety									
Variance accounted for	17.5%	8.4%	7.9%	20.7%	12.7%	6.8%	29.0%	12.6%	7.3%

Note. +/− indicates whether effect is positive or negative; *$p < .01$, **$p < .001$.

streams. Combined with UGT, these results suggest people spend more time watching live-steams to meet their need for social interaction. This is consistent with recent research finding a significant correlation between motivation for socialization and usage time (Gros, Wanner, Hackenholt, Zawadzki, & Knautz, 2017). However, while previous research has identified a significant correlation between usage time and motivation to use Twitch to be a part of a community (Gros et al., 2017), sense of community was not significantly associated with time spent watching live-streams in this study. Additionally, outside the realm of streaming, social motivations have been shown to impact usage in contexts such as social media (Papacharissi & Mendelson, 2011; Whiting & Williams, 2013). Also, external support was negatively associated with the increase of time, with decreased external support associated with greater time spent in live-streams. This suggests individuals with fewer social ties in real life are more motivated to spend time engaging in live streams, than people with greater social ties. This supports research that found people spend more time in online interactions to compensate for a lack of social interactions in their lives (Desjarlais & Willoughby, 2010; Miller, 2011). This is also consistent with UGT, which maintains viewers will participate in a way that fulfils their needs. However, it is also possible that individuals' lacking social ties, may have more free time to spend online, accounting for increased time spent watching live-streams. Both theories may be applicable, with individuals driven by different motivations. Therefore, further research into this area is recommended.

Twitch live-stream subscriptions and donations, which involve financial contribution, were motivated by desire for social interaction and a sense of community. This provides insight for streamers who want to develop their live-stream community. Sense of community can be fostered by providing viewers with membership, influence, fulfilment of needs, and emotional connection (Hamilton et al., 2014; McMillan & Chavis, 1986). Therefore, increasing participation in a channel may be strengthened by encouraging viewers to vote in polls to choose the game you play, including viewers in decision making and planning streams, noticing and greeting them personally, and generally fostering socialising in the community. These findings are consistent with previous research that found that individuals are motivated to subscribe in order to build deeper involvement with the community and feel like a larger part of shared experiences (Sjöblom & Hamari, 2017). It is also consistent with data gathered from an online music consumption platform, finding that willingness to pay for premium services was more strongly associated to community participation than to content consumption (Oestreicher-Singer & Zalmanson, 2012). While motivations to donate on Twitch have not previously been explored, the findings are in line with other research finding social incentives for donating money, time, and blood (Bekkers, 2010; Lee, Piliavin, & Call, 1999; Satow, 1975). This study builds and contributes to current live-stream research by providing further insight into subscription motivations, while also identifying the motivations to donate for the first time.

Overall, the results show that six out of the eight factors in the proposed socio-motivation model partly explained live-stream emotional attachment and behavioural engagement. A desire for social interaction and a sense of belonging to an online community were the most consistent and strongest motivators of live-stream engagement. In contrast, a desire for social support or compensation for social anxiety did not explain engagement. These results are contrary to previous research identifying social support and social anxiety as factors increasing participation in online communities (Mazalin & Klein, 2008; Valkenburg & Peter, 2009), suggesting that motivations for live-stream engagement may differ from other online communal activities. While Twitch primarily operates as a live-streaming platform for gaming content, this study suggests that the motivations to watch games differ from the motivations to play. Although this study finds social interaction and sense of community the most consistent motivators for live-stream engagement, a recent meta-analysis of 48 studies showed that enjoyment and usefulness were the most important determinants

of game use (Hamari & Keronen, 2017).

The stronger association between social motivators and stream engagement for viewers who preferred smaller channels seems to confirm that smaller channels provide richer social interactions and a greater sense of community than larger channels, which is consistent with past qualitative research (Hamilton et al., 2014). Live-streams of all sizes function as socially interactive and community-oriented environments, however smaller channels can offer viewers experiences that a larger channel cannot. Streams with smaller audiences can greet viewers by name, and provide direct acknowledgement and appreciation of their presence. Smaller channels can offer audiences deeper interactive involvement and stronger influence over the live-stream content. It is easier for viewers in smaller channels to communicate with one other, thus fostering a community where they know each other by their usernames. These findings suggest an interesting dilemma for the hosts of live-streaming channels; on one hand it may be beneficial to run a small channel but, on the other hand, a successful small channel will eventually grow and potentially lose its high communal appeal.

4.1. Limitations, future directions, and implications

This research offers new insight into the motivations of live-stream viewers, however the results should be interpreted with some caution. As a cross-sectional, single-source, self-report survey-based study, findings may be subject to common method bias (Podsakoff et al., 2003). However, self-report measures were considered to be the most accurate and readily available way of assessing motivations for the current study, and a Harman's single factor test suggested the data was free from common method bias (Podsakoff et al., 2003). Although motivations can be unconscious, from a UGT perspective, consumers are conscious and selective about their media consumer choices in order to satisfy their needs.

The internal consistency of the measures used in the current study was reasonable, however there is clearly also need for further psychometric development of the socio-motivational measures. Future studies could consider further developing, and testing, a six-factor model of live-stream motivations (i.e., dropping social anxiety and social support because they were not significant predictors of live-stream engagement). Measurement of these motivations could be improved by expanding the number of items and including some negatively worded items. Content, construct, and criterion validity of the socio-motivation measures could then be tested with a view to improving confidence in its utility for hypothesis testing.

The current study's cross-sectional designs do not enable causal relationships to be inferred, so further research should also consider experimental and longitudinal designs. The bulk of participants were conveniently sourced from Reddit, therefore the generalisability of findings may be limited. Twitch users who use Reddit are likely to be more socially engaged in the internet and thus are not necessarily representative of all Twitch viewers. Similarly, while 75% of Twitch viewers are male (Twitch, 2016), 95.6% of respondents in this study were male. This heavily male gender distribution is consistent with other estimates of gender distribution among video game stream users (e.g., Quantcast, 2016). Future research could benefit from seeking a more gender-representative sample and consider gender differences in socio-motivations for engaging in live-stream participation. Another consideration is that respondents' country of origin was not known. Future research could collect such data and investigate cultural differences in motivations of live-stream engagement.

The socio-motivational model strongly explained live-stream engagement for viewers who prefer small channels, with small to moderate explanation of engagement in larger channels. As larger channel participation represents the majority of live-stream use (Kaytoue, Silva, Cerf, Meira Jr, & Raïssi, 2012) further research is needed to better understand engagement in these environments. Comparing the explanatory power of the socio-motivational model by channel size provided useful insight into how motivations are likely to vary by stream type. Future research could delve further into socio-motivational differences between other live-stream classifications, such as by genre (e.g., eSports versus variety streamers) and/or viewer type (e.g., personality) (Sjöblom et al., 2017).

Overall, this study has illustrated the potential utility of applying a UGT framework to understanding live-stream engagement. Different viewer motivations appear to guide participants towards different types of live-stream communities and types of engagement. While the findings seem to support the potential of a UGT approach, the extent to which motivations were gratified was not explored. Such research would be an important next step in understanding the motivational dynamics of live-stream community participation from a functional perspective.

Understanding why millions of people spend so much time and money, watching, subscribing, and donating to live-streamers has several practical implications. Motivations to engage in live-stream entertainment have a strong social and community emphasis compared to engagement in mass media such as television. Live-streamers wanting to grow their viewer base should explore how to better satisfy their viewers' needs, particularly social needs. Active social streams which cultivate a strong sense of community and belonging may use techniques such as viewer polls to help make decisions, giving viewers' attention (e.g., greeting them personally, acknowledging and showing appreciation of their participation), and enhancing membership identity (e.g., providing additional social perks for subscribers).

Spending time on the internet can detract from people's social lives (Kim, LaRose, & Peng, 2009), however the current study suggests that certain types of internet activity, such as engaging in live-streams, can provide an alternate form of community and social engagement. These findings may also encourage families and friends to recognise the social benefits of live-stream engagement. These results help to explain why the popularity of live-streaming is growing rapidly; consumers are motivated to engage not just for entertainment and information, but also for social interaction and belonging.

5. Conclusions

This study investigated how motivations of users of live-streams explain their psychological and behavioural engagement with live-stream media. Socially and community-motivated engagement was associated with greater emotional connection, time spent, and financial contribution to live-streams, particularly for smaller channels. Engagement with larger, more populated channels was less socially driven. Regardless, live-stream engagement across all channel sizes was motivated by a desire for social interaction and sense of community, illustrating social relations as a key motivational force for live-stream engagement. This study has demonstrated the potential utility of a functional approach to understanding the relationship between viewer motivations and their psychological and behavioural live-stream experience, and helps to explain why millions of people are spending large parts of their lives engaged in live-stream communities.

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References

Bargh, J. A., & McKenna, K. Y. (2004). The internet and social life. *Annual Review of Psychology, 55*, 573–590. https://doi.org/10.1146/annurev.psych.55.090902.141922.

Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin, 117*, 497–529. https://doi.org/10.1037/0033-2909.117.3.497.

Bekkers, R. (2010). Who gives what and when? A scenario study of intentions to give time and money. *Social Science Research, 39*(3), 369–381. https://doi.org/10.1016/j.ssresearch.2009.08.008.

Brandtzæg, P. B., & Heim, J. (2009). Why people use social networking sites. *Online Communities and Social Computing, 5621*, 143–152. https://doi.org/10.1007/978-3-642-02774-1_16.

Cha, J. (2014). Usage of video sharing websites: Drivers and barriers. *Telematics and Informatics, 31*, 16–26. https://doi.org/10.1016/j.tele.2012.01.003.

Chang, Y. P., & Zhu, D. H. (2011). Understanding social networking sites adoption in China: A comparison of pre-adoption and post-adoption. *Computers in Human Behavior, 27*(5), 1840–1848. https://doi.org/10.1016/j.chb.2011.04.006.

Chen, C. C., & Lin, Y. C. (2018). What drives live-stream usage intention? The perspectives of flow, entertainment, social interaction, and endorsement. *Telematics and Informatics, 35*(1), 293–303. https://doi.org/10.1016/j.tele.2017.12.003.

Cheung, G., & Huang, J. (2011). Starcraft from the stands: Understanding the game spectator. In *Proceedings of the ACM SIGCHI conference on human factors in computing systems, USA* (pp. 763–772). https://doi.org/10.1145/1978942.1979053.

Chiang, H. S., & Hsiao, K. L. (2015). YouTube stickiness: The needs, personal, and environmental perspective. *Internet Research, 25*, 85–106. https://doi.org/10.1108/IntR-11-2013-0236.

Chin, W. W. (1998). The partial least squares approach to structural equation modeling. In G. A. Marcoulides (Ed.), *Modern methods for business research* (pp. 295–358). Mahwah, NJ: Lawrence Erlbaum Associates.

Chiu, C. M., Hsu, M. H., & Wang, E. T. (2006). Understanding knowledge sharing in virtual communities: An integration of social capital and social cognitive theories. *Decision Support Systems, 42*(3), 1872–1888. https://doi.org/10.1016/j.dss.2006.04.001.

Cohen, S., & Hoberman, H. M. (1983). Positive events and social supports as buffers of life change stress. *Journal of Applied Social Psychology, 13*, 99–125. https://doi.org/0.1111/j.1559-1816.1983.tb02325.x.

Desjarlais, M., & Willoughby, T. (2010). A longitudinal study of the relation between adolescent boys and girls' computer use with friends and friendship quality: Support for the social compensation or the rich-get-richer hypothesis. *Computers in Human Behavior, 26*, 896–905. https://doi.org/10.1016/j.chb.2010.02.004.

Dunbar, R. I. (1992). Neocortex size as a constraint on group size in primates. *Journal of Human Evolution, 22*(6), 469–493. https://doi.org/10.1016/0047-2484(92)90081-J.

D'onfro, J. (2015, December 4). *Facebook takes on streaming apps live Periscope and Meerkat with its new 'live video' feature. Business Insider*. Retrieved from http://www.businessinsider.com.au/facebook-introduces-live-video-feature-2015-12.

Ellison, N. B., Steinfield, C., & Lampe, C. (2007). The benefits of Facebook "friends": Social capital and college students' use of online social network sites. *Journal of Computer-mediated Communication, 12*, 1143–1168. https://doi.org/10.1111/j.1083-6101.2007.00367.x.

Fitzgerald, D., & Wakabayashi, D. (2014, February 3). Apple quietly builds new networks. *The Wall Street Journal*. Online. Retrieved from http://www.wsj.com/articles/SB10001424052702304851104579361201655365302.

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research, 18*(1), 39–50. https://doi.org/10.2307/3151312.

Gamezone. (2014, May 15). *The rise of Twitch.tv: How this giant came to be. Gamezone*. Retrieved from http://www.mweb.co.za/games/view/tabid/4210/Article/13210/The-rise-of-Twitchtv-How-this-giant-came-to-be.aspx.

Gros, D., Wanner, B., Hackenholt, A., Zawadzki, P., & Knautz, K. (2017). World of streaming. Motivation and gratification on Twitch. In G. Meiselwitz (Ed.), *Social computing and social media. Human behavior* (pp. 44–57). Cham, Switzerland: Springer. https://doi.org/10.1007/978-3-319-58559-8_5.

Hamari, J., & Keronen, L. (2017). Why do people play games? A Meta-Analysis. *International Journal of Information Management, 37*(3), 125–141. https://doi.org/10.1016/j.ijinfomgt.2017.01.006.

Hamari, J., & Sjöblom, M. (2017). What is eSports and why do people watch it? *Internet Research, 27*(2). https://doi.org/10.1108/IntR-04-2016-0085.

Hamilton, W. A., Garretson, O., & Kerne, A. (2014). Streaming on Twitch: Fostering participatory communities of play within live mixed media. In *Proceedings of the 32nd annual ACM conference on Human factors in computing systems, USA* (pp. 1315–1324). https://doi.org/10.1145/2556288.2557048.

Hanson, G., & Haridakis, P. (2008). YouTube users watching and sharing the news: A uses and gratifications approach. *Journal of Electronic Publishing, 11*(3). https://doi.org/10.3998/3336451.0011.305.

Jöreskog, K. G., & Sörbom, D. (1996). *LISREL 8: User's reference guide*. Chicago, USA: Scientific Software International.

Kaytoue, M., Silva, A., Cerf, L., Meira, W., Jr., & Raïssi, C. (2012). Watch me playing, I am a professional: A first study on video game live streaming. In *Proceedings of the 21st international ACM conference companion on world wide web, New York, NY, USA* (pp. 1181–1188). https://doi.org/10.1145/2187980.2188259.

Kim, J., LaRose, R., & Peng, W. (2009). Loneliness as the cause and the effect of problematic internet use: The relationship between Internet use and psychological well-being. *CyberPsychology and Behavior, 12*(4), 451–455. https://doi.org/10.1089/cpb.2008.0327.

Kim, Y., Sohn, D., & Choi, S. M. (2011). Cultural difference in motivations for using social network sites: A comparative study of American and Korean college students. *Computers in Human Behavior, 27*(1), 365–372. https://doi.org/10.1016/j.chb.2010.08.015.

Ku, Y. C., Chen, R., & Zhang, H. (2013). Why do users continue using social networking sites? An exploratory study of members in the United States and Taiwan. *Information & Management, 50*, 571–581. https://doi.org/10.1016/j.im.2013.07.011.

Lee, L., Piliavin, J. A., & Call, V. R. (1999). Giving time, money, and blood: Similarities and differences. *Social Psychology Quarterly, 62*(3), 276–290. https://doi.org/10.2307/2695864.

Leslie, C. (2015, January 30). *100 million people are watching Twitch every month*. The Daily Dot. Retrieved from http://www.dailydot.com/esports/twitch-2014-year-in-review/.

Mattick, R. P., & Clarke, J. C. (1998). Development and validation of measures of social phobia scrutiny fear and social interaction anxiety. *Behaviour Research and Therapy, 36*, 455–470. https://doi.org/10.1016/S0005-7967(97)10031-6.

Mazalin, D., & Klein, B. (2008). Social anxiety and the internet: Positive and negative effects. *Sensoria: A Journal of Mind, Brain & Culture, 4*(2), 43–50. https://doi.org/10.7790/ejap.v4i2.137.

McMillan, D. W., & Chavis, D. M. (1986). Sense of community: A definition and theory. *Journal of Community Psychology, 14*, 6–23. https://doi.org/10.1002/1520-6629.

Miller, V. (2011). *Understanding digital culture*. Kent, UK: Sage Publications. Retrieved from https://au.sagepub.com/en-gb/oce/understanding-digital-culture/book232956.

Needleman, S. E. (2015, January 29). *Twitch's viewers reach 100 million a month*. Retrieved June 6th, 2016, from The Wall Street Journal http://blogs.wsj.com/digits/2015/01/29/twitchs-viewers-reach-100-million-a-month/.

Nunnally, J. C. (1978). *Psychometric theory*. New York: McGraw-Hill.

O'Neill, P. H. (2014, January 16). *Twitch dominated streaming in 2013, and here are the numbers to prove it*. The Daily Dot. Retrieved from http://www.dailydot.com/esports/twitch-growth-esports-streaming-mlg-youtube-2013/.

Oestreicher-Singer, G., & Zalmanson, L. (2012). Content or community? A digital business strategy for content providers in the social age. *MIS Quarterly, 37*(2), 591–616. https://doi.org/10.2139/ssrn.1536768.

Papacharissi, Z., & Mendelson, A. (2011). Toward a new(er) sociability: Uses, gratifications and social capital on Facebook. In *Media perspectives for the 21st century* (pp. 212–230). New York, NY: Routledge.

Peterson, N. A., Speer, P. W., & McMillan, D. W. (2008). Validation of a brief sense of community scale: Confirmation of the principal theory of sense of community. *Journal of Community Psychology, 36*, 61–73. https://doi.org/10.1002/jcop.20217.

Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioural research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903. http://doi.org/10.1037/0021-9010.88.5.879.

Quantcast. (2016). *Twitch.tv traffic and demographics*. Retrieved November 15th 2016 from https://www.quantcast.com/twitch.tv#/demographicsCard.

Ridings, C. M., & Gefen, D. (2004). Virtual community attraction: Why people hang out online. *Journal of Computer-mediated Communication, 10*(1). https://doi.org/10.1111/j.1083-6101.2004.tb00229.x.

Rodebaugh, T. L., Woods, C. M., & Heimberg, R. G. (2007). The reverse of social anxiety is not always the opposite: The reverse-scored items of the Social Interaction Anxiety Scale do not belong. *Behavior Therapy, 38*(2), 192–206. https://doi.org/10.1016/j.beth.2006.08.001.

Rubin, A. M. (2009). Uses and gratifications. In *The SAGE handbook of media processes and effects*. California, CA: Sage Publications.

Ruggiero, T. E. (2000). Uses and gratifications theory in the 21st century. *Mass Communication & Society, 3*, 3–37. https://doi.org/10.1207/S15327825MCS0301_02.

Satow, K. L. (1975). Social approval and helping. *Journal of Experimental Social Psychology, 11*(6), 501–509. https://doi.org/10.1016/0022-1031(75)90001-3.

Shao, G. (2009). Understanding the appeal of user-generated media: A uses and gratification perspective. *Internet Research, 19*(1), 7–25. https://doi.org/10.1108/10662240910927795.

Sjöblom, M., & Hamari, J. (2017). Why do people watch others play video games? An empirical study on the motivations of Twitch users. *Computers in Human Behavior, 30*(1), 1–12. https://doi.org/10.1016/j.chb.2016.10.019.

Sjöblom, M., Törhönen, M., Hamari, J., & Macey, J. (2017). Content structure is king: An empirical study on gratifications, game genres and content type on Twitch. *Computers in Human Behavior, 73*, 161–171. https://doi.org/10.1016/j.chb.2017.03.036.

Twitch. (2016). *Audience*. Retrieved from http://twitchadvertising.tv/audience/.

Twitch. (2017). *Twitch 2016 retrospective*. Retrieved from http://www.twitch.tv/year/2016.

Valkenburg, P. M., & Peter, J. (2009). Social consequences of the internet for adolescents: A decade of research. *Current Directions in Psychological Science, 18*, 1–5. https://doi.org/10.1111/j.1467-8721.2009.01595.x.

West, R. L., & Turner, L. H. (2010). *Introducing communication theory: Analysis and application*. New York, NY: McGraw-Hill Humanities/Social Sciences/Languages. Retrieved from http://digitalcommons.usm.maine.edu/facbooks/127/.

Whiting, A., & Williams, D. (2013). Why people use social media: A uses and gratifications approach. *Qualitative Market Research: An International Journal, 16*, 362–369. https://doi.org/10.1108/QMR-06-2013-0041.

Wilhelm, A. (2013, March 30). *As DailyMotion and Youtube turn up the pressure, Twitch looks to retain livestreaming ascendance*. The Next Web. Retrieved from http://thenextweb.com/insider/2013/03/30/as-dailymotion-and-youtube-turn-up-the-pressure-twitch-looks-to-retain-livestreaming-ascendance/.

Zhu, D. H., & Chang, Y. P. (2014). Understanding motivations for continuance intention of online communities in China: A comparison of active users of social networking sites and virtual communities. *Information Development, 30*, 172–180. https://doi.org/10.1177/0266666913482597.